Abraham I. George

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https://scholar.google.com/citations?user=Bxt7N18AAAAJ&hl=en

EDUCATION

Ph.D. in Mechanical Engineering
Carnegie Mellon University, Pittsburgh PA – Defended November 2025 GPA: 4.00
Area of Research: Machine learning for complex robotic manipulation tasks, with an emphasis on efficient learning from human demonstrations

Master of Science in Mechanical Engineering Advanced Studies – Robotics
Carnegie Mellon University, Pittsburgh PA GPA: 4.00
Graduated in the Class of 2022

Bachelor of Science in Mechanical Engineering
The Pennsylvania State University, University Park PA GPA: 4.00
Graduated Summa Cum Laude in the Class of 2021

Relevant Graduate Coursework
Deep Learning Systems, Deep Reinforcement Learning, Engineering Computation, Modern Control Theory, Multivariable Linear Control, Control Systems Integration, Sensing and Sensors, Robotic Dynamics and Analysis

TECHNICAL SKILLS

Programming	Python, C/C++, C#, MATLAB, Bash
ML & Deep Learning	PyTorch, Tensorflow, CUDA, Hugging Face
Other Software	ROS, ROS2, Unity, SolidWorks, Simulink, PyBullet, Mujoco, Issac Lab

RESEARCH EXPERIENCE

Tactile Sensing for Robot Learning
- Used visuo-tactile pretraining to improve imitation learning on dexterous manipulation tasks [1, 7]. Pretraining moderately improved visuo-tactile agents and significantly improved vision-only agents, achieving performance on-par with visuo-tactile agents without tactile sensing at inference.
- Developed a novel tactile sensor for robotic applications, designed to be inexpensive and durable [2].

Data Efficiency in Robot Learning via Automated Demonstration Generation
- Developed a system to generate demonstrations by augmenting a single human demo.
- Using this augmentation method on a single human demonstration, was able to significantly improve the training speed of an RL agent and allow it to solve a previously unsolved task [3].
- Extended this augmentation method to imitation learning (IL), where it was used to train an agent from a single human demonstration [4].
- Incorporated LLMs to assist in data generation (with LLM optimization), both to train IL agents and to create open-loop control policies (which can be combined with a trained IL agent for close-loop control) [5].
- Leveraged VLA encoders and a large-scale robot database to create a high performing NN based IL policy. [8]

Teleoperation For Robot Control
- Developed an open-source VR robot teleoperation system to allow for intuitive demonstration collection [6]. The system creates a digital twin of the robot's workspace and supports teleoperation of anthropomorphic hands.

Leveraging Foundation Models for Robotic Control
- Developed PLATO, a robotic planning system using LLMs to understand natural language instructions, predict tool affordances, and execute complex tasks in unstructured environments without predefined knowledge [9].
- Made a hybrid planning framework that integrates LLM semantic reasoning with A* on a low-cost robot [10].
- Combined modular construction with LLM-based planning for robust aerial additive manufacturing [11].

WORK EXPERIENCE

Mitsubishi Electric Research Lab (MERL) Internship: May-Sept 2025
Conducted research into deep learning for human-robot collaboration, which was submitted to ICRA 2026.

LEADERSHIP EXPERIENCE

Cluster Management
Managed the computational clusters for the lab, total value ~$300,000. In charge of managing user access, balancing computation load, ensuring compatibility, and general debugging.

Robotics Group Lead
Lead the robotics subgroup in the lab, consisting of ~10 master's students. Duties included formulating research projects, providing feedback on progress, giving technical assistance, and leading a weekly group meeting.

PROJECTS

Bio-Inspired Jumping Robot Leg to Study the Effects of Leg Morphology
Created a robotic leg to study the effects of leg morphology on jumping performance. Our work showed the effect of altering different bone segments in the leg on total vertical jump height.
(https://www.youtube.com/watch?v=Z9h7IUoEveg)

CrazyFlie Goalie
Developed and implemented a gain scheduled LQR controller for object interception on a CrazyFlie drone. The drone intercepted thrown 1cm balls in the air, successfully guarding a 1-meter square goal.

Senior Design Project – Ship Deck Simulator for JHU APL
Designed a platform to emulate the behavior of a ship's deck in various sea conditions to help John Hopkins University Applied Physics Lab test navel UAVs. 2nd Place winner (out of 130) of the PSU 2021 Senior Design Showcase. (https://sites.psu.edu/lfshowcasesp21/2021/04/28/shipboard-launch-and-recovery-simulator/)

SELECTED PUBLICATIONS

[1] **George, A**., Gano, S., Katragadda, P., & Farimani, A. B. (2025). VITaL Pretraining: Visuo-Tactile Pretraining for Tactile and Non-Tactile Manipulation Policies. *ICRA 2025*
[2] **George, A.**, Chen, Y., Dikshit, A., Pak, P., & Farimani, A. B. (2024). BeadSight: An Inexpensive Tactile Sensor Using Hydro-Gel Beads. *IEEE Sensors*
[3] **George, A.**, Bartsch, A., & Farimani, A. B. (2023). Minimizing human assistance: Augmenting a single demonstration for deep reinforcement learning. *ICRA 2023*
[4] **George, A.**, & Farimani, A. B. (2023). One ACT Play: Single Demonstration Behavior Cloning with Action Chunking Transformers. *arXiv:2309.10175*
[5] **George, A.**, & Farimani, A. B. (2025). LLM Trainer: Automated Robotic Data Generating via Demonstration Augmentation using LLMs. *Under Review, ICRA 2026*
[6] **George, A.**, Bartsch, A., & Farimani, A. B. (2025). OpenVR: Teleoperation for manipulation. *SoftwareX*
[7] Gano, S., **George, A.**, & Faramani, A. B. (2025). Low Fidelity Visuo-Tactile Pretraining Improves Vision-Only Manipulation Performance. *IROS 2025*
[8] Kwon, O., **George, A.**, Bartsch, A., & Faramani A. B. (2025). RT-cache: Efficient Robot Trajectory Retrieval System. *Humanoids 2025.*
[9] Car, A., Yarlagadda, S. S., Bartsch, A., **George, A.**, & Farimani, A. B. (2024). PLATO: Planning with LLMs and Affordances for Tool Manipulation. *Under Review RA-L*
[10] Barkley, J., George, A., & Faramani A. B. (2025). Semantic Intelligence: Integrating GPT-4 with A Planning in Low-Cost Robotics. *Under Review, ICRA 2026*
[11] Merrill, C., Raman, **George, A.**, & Faramani A. B. (2025). LLM-drone: aerial additive manufacturing with drones planned using large language models. *Construction Robotics*

HONORS

PSU President Sparks Award
Joseph B. Wharton Memorial Scholarship
The Evan Pugh Scholar Award